Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

2.2 – Distance–Time Graphs: Match the Graph

Unit 2: Forces and Motion

Warm-up

A person runs **250 m** in **30 seconds**. What is their speed? Give your answer to 1 decimal place.

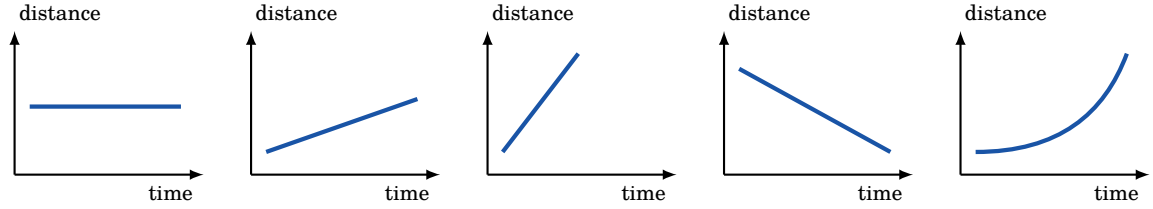
speed = distance ÷ time = _____ m ÷ _____ s = _____ m/s

Check: speed × time gives you back 250 m.

Reading a distance–time graph

Distance–time graph a graph with distance up the side and time along the bottom. Its shape tells the story of how something moved.

Today the distance is **your distance from a motion sensor**. Walk **away** from the sensor and the line goes **up**. Walk **towards** it and the line goes **down**. Stand still and the line stays **flat**.

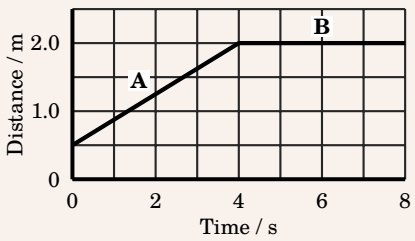


Stationary	**Constant speed**	**Faster**	**Constant speed**	**Speeding up**
standing still	away from it	steeper line	towards it	line curves up

One idea explains every shape: **the steepness of the line shows the speed.** Whatever the speed does, the steepness does too.

- A **flat line** has no steepness: no speed, so **stationary**.
- A **straight line** keeps the same steepness, so the speed is **constant**. The **steeper** the line, the **faster** the speed.
- A **curved line** changes steepness, so the speed is **changing**: getting steeper is speeding up, getting flatter is slowing down.
- **Up or down** shows the direction: up is **away from** the sensor, down is **towards** it.

Worked example — read the story



Part A is a straight line going up: the walker moved **away** from the sensor at a **constant speed**, from 0.5 m to 2.0 m in 4 s.

Part B is flat: the walker was **stationary**, standing 2.0 m from the sensor for the last 4 s.

Match the Graph — the method

Walk the graph

Kit: a PASCO motion sensor on a laptop running SPARKvue, pointing along a clear strip of floor. Your teacher sets it up so the screen shows a graph of **position** (your distance from the sensor, in metres) against **time**, on the same axes as the graphs overleaf.

Steps

1. **Plan first.** Read the graph, and decide where to start standing and how to move. Use the numbers: a line starting at 1.5 m means you start 1.5 m from the sensor.
2. **Walk it.** Stand at your starting distance, facing the sensor. On **Start**, move the way you planned. The line on the screen follows you.
3. **Compare.** Hold your line up against the graph on the sheet. Adjust the plan and walk it again until the two match.
4. **Write it down.** Under the graph, describe how you moved, using the words in the box below. Tick *Matched it?* when your line fits.

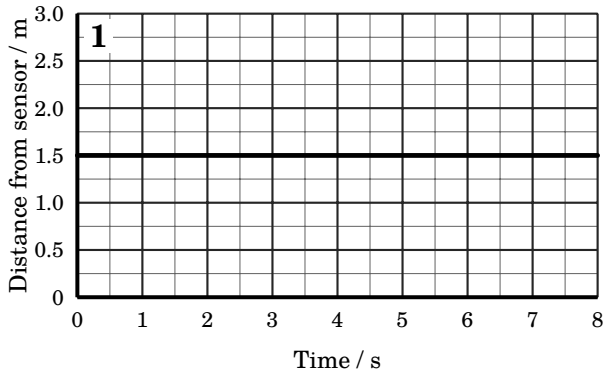
Check: walking away makes your line climb; walking towards makes it fall.

Care: keep the strip in front of the sensor clear of bags and chairs, and walk — the sensor reads the first thing in front of it.

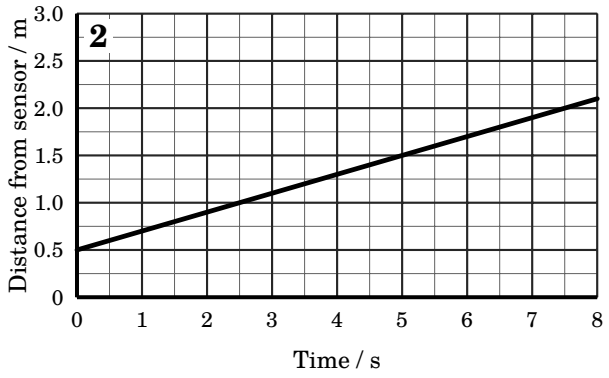
Words to use: stationary • constant speed • away from • towards • faster • slower • speeding up • slowing down

The six graphs

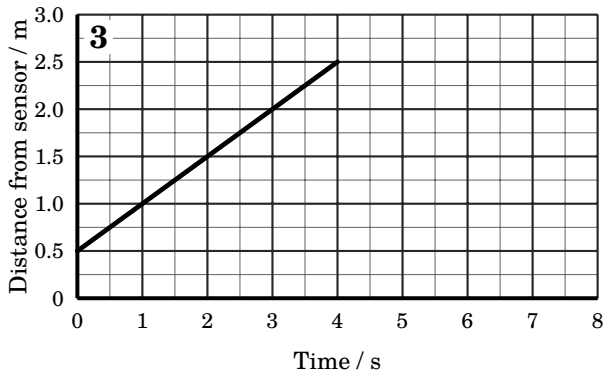
Work through them in order. Number 5 has four parts — describe each one.



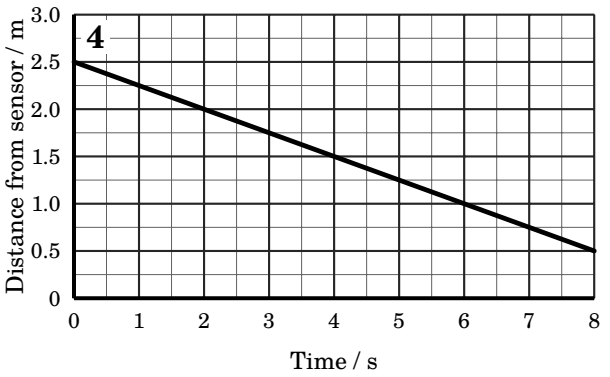
How I moved: **Matched it?** ☐



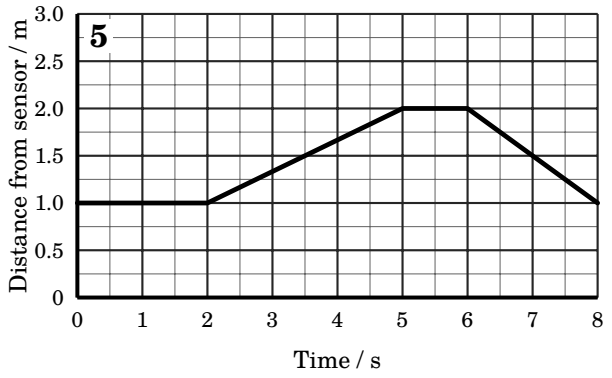
How I moved: **Matched it?** ☐



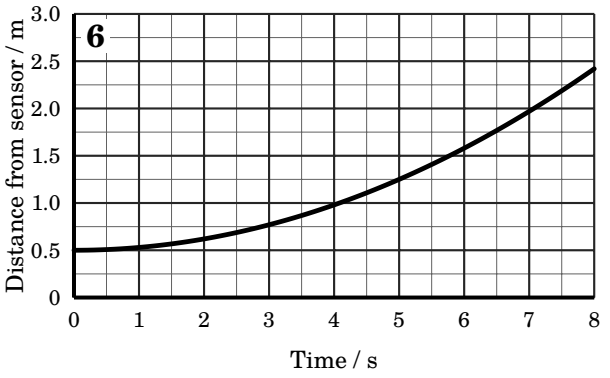
How I moved: **Matched it?** ☐



How I moved: **Matched it?** ☐



How I moved: **Matched it?** ☐



How I moved: **Matched it?** ☐

What the graphs show

1. On a distance–time graph, a flat line means the walker is _____.

2. Graphs 2 and 3 are both straight lines going up. Which one was the faster walk, and how can you tell from the graph?

_____

_____

3. Use Graph 3 to work out the walker's speed. They moved from 0.5 m to 2.5 m in 4 s.

speed = distance ÷ time = _____ m ÷ _____ s = _____ m/s

Check: the distance is how far they *moved*, end minus start.

To go further — the impossible graph

Nobody in the room can walk a graph that goes **straight up**, a vertical line. Explain why, using what a vertical line would mean about distance and time.

_____

_____

_____